

finding a 'difference' between two binary files is not simple and straightforward. Ideally, version control software, when used, should be hosted on a separate server, which is not an option at home. Additionally, maintaining the repository on an external HDD was next to impossible.

Then I started searching for personal backup software that would allow me to maintain backups of data and that, too, on external media like a portable HDD. I tried multiple solutions, two of them being Microsoft SyncToy and Cobian Backup. While I liked each of these tools (with a preference for Cobian Backup), I found that the GUI interface, and the way these tools stored the backup commands, created a problem that needed additional effort each time I wished to use the tools for backup.

While using these tools, I faced a problem with the backup commands due to the way Windows uses external media. Whenever external media is connected to a computer, there is no guarantee that it will be assigned the same drive letter as the one assigned the last time the media was connected. For example, if the connected media is assigned the drive letter *H:* today, there is no guarantee that it will be assigned the same drive letter tomorrow. If you have other media already connected, the portable HDD will get assigned the next available drive letter. As the backup commands configured in these tools refer to an absolute path that includes the drive letter, the command does not work if the drive letter is different. If the external HDD is assigned a different drive letter, you will have no choice but to edit the backup command to point it to the present drive letter. Now imagine the effort needed if such changes have to be applied to multiple backup commands, each time you wish to take a backup.

While searching for more tools, I came across a simple backup tool, namely, Sync (alternately named Syncdir).

Introducing Sync

Sync is a Java application that needs to be invoked from the

```
C:\Windows\system32\cmd.exe
Could Not Find ST:\BCHRP\log\doc.log
Sync 2.1 Copyright 2007 Zach Scrivena 2007-12-09
DIRECTORY SYNCHRONIZATION
Log file: "E:\BCHRP\log\doc.log"
Source directory: "D:\doc\"
Target directory: "E:\BCHRP\doc\"
File-matching attributes: {name,size,time,crc}
Source file/directory filter: NOT glob(".*")
Target file/directory filter: NOT glob(".*")
Filter mode: filename
SUBDIRECTORIES: ".\"
No. of source files matched: 0 of 1
No. of unmatched source files to be copied: 1
[C1] "CSI Journal Vol 2 No.4 2015.pdf" (4.4 Mb)
SYNCHRONIZATION REPORT
No. of source subdirectories scanned : 1
No. of source files scanned : 1
No. of target files scanned : 0
No. of source files matched [N] : 0
No. of unmatched source files [C] : 1 (1 copied)
No. of unmatched target files/directories [D] : 0 (0 deleted)
Sync is done!
Backup for doc done.
Press any key to continue . . .
```

Figure 1: Taking backups for the first time

command line. I can almost hear you grumble: "Ew! The command line. How can you even recommend a command line application in these days of GUI systems?" But, in my experience, this command line interface provides me the maximum flexibility.

On my local machine, I always have a JVM instance available either as a JDK or a JRE, along with a properly configured PATH variable and a JAVAHOME variable. To make the task of backing up easier, I have created batch files that contain the backup command (using Sync). The batch files are then stored on the portable device. Thus, whenever I wish to take a backup, I simply connect the portable device and execute the backup script. As the source directory does not change (being local storage) and the destination directory is specified using a relative path, the batch scripts do not need to be edited – irrespective of what drive letter is assigned to it. When launched, the script executes in a dedicated command prompt and I can continue with other activities. I only need to monitor the progress once in a while. Once the backup is complete, I go through the log file to check for errors and my backup work is done.

By using a Windows batch file, my backup activity is reduced to a double click or an 'Enter' key-press, which is all that's needed to launch the script. The scripts have made the task so simple that even after maintaining two backup copies of my personal photographs, email and other documents, the only trouble I face is that of maintaining a regular schedule to keep the backup copies suitably fresh.

Using Sync

To use Sync, the command format is as follows:

```
java -jar Sync.jar <switches> ["Source"] ["Target"]
```

Using this command synchronises the ["Target"] to match the ["Source"]. It should be noted that only the

```
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Sync 2.1 Copyright 2007 Zach Scrivena 2007-12-09
DIRECTORY SYNCHRONIZATION
Log file: "E:\BCHRP\log\doc.log"
Source directory: "D:\doc\"
Target directory: "E:\BCHRP\doc\"
File-matching attributes: {name,size,time,crc}
Source file/directory filter: NOT glob(".*")
Target file/directory filter: NOT glob(".*")
Filter mode: filename
SUBDIRECTORIES: ".\"
No. of source files matched: 1 of 4
[Hit enter] "CSI Journal Vol 2 No.4 2015.pdf"
No. of unmatched source files to be copied: 3
[C1] "B Comparison of Opensource ERP systems.pdf" (1.1 Mb)
[C2] "Open Source Software IP Risk audits.pdf" (30.6 kb)
[C3] "MITIpaper2.pdf" (1.1 Mb)
SYNCHRONIZATION REPORT
No. of source subdirectories scanned : 1
No. of source files scanned : 4
No. of target files scanned : 1
No. of source files matched [N] : 1
No. of unmatched source files [C] : 3 (3 copied)
No. of unmatched target files/directories [D] : 0 (0 deleted)
Sync is done!
Backup for doc done.
```

Figure 2: Updating the backup. New files added